Region-Based Progressive Alignment Network for Multimodal Remote Sensing Object Detection

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***Abstract*—Weakly misalignment problem severely degrades multimodal remote sensing object detection, particularly when the displacement is spatially nonuniform and small objects are involved. To address these issues, we propose a region-based progressive alignment network (RPANet) through regional explicit–implicit alignment, realizing reliable local offset estimation and offset-guided implicit alignment. Specifically, we propose a redundancy-aware regional offset modeling module (R^2OMM) to estimate reliable regional offsets, where the regional spatial modulation (RSM) enhances informative spatial structures within local windows, and the channel-recalibrated offset modeling (COM) suppresses redundant responses while predicting explicit regional offsets for coarse geometric correction. In addition, we design a dual-path synergistic implicit alignment module (DPSIAM) to perform offset-guided implicit alignment by jointly modeling intramodal refinement and intermodal interaction. Experiments on the DroneVehicle, VEDAI, DVTOD, and LLVIP datasets verify the effectiveness of RPANet and show that it achieves state-of-the-art detection performance, especially under weakly misaligned scenarios.**

***Index Terms*—Multimodal object detection, progressive alignment, regional offset estimation, remote sensing, Weakly misalignment problem.**

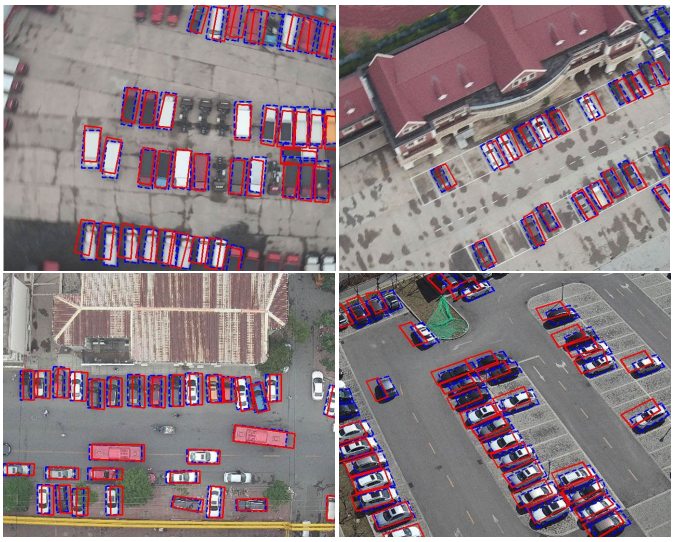


Fig. 1. Visualization of weakly misalignment, highlighting the regional differences and positional shifts of small objects. The blue boxes indicate annotations in the RGB modality, and the red boxes represent annotations in the IR modality.

I. Introduction

As ONE of the core tasks in Earth observation technology, remote sensing object detection plays an irreplaceable role in fields such as national defense security, urban planning, environmental monitoring, and disaster assessment [1], [2], [3]. Traditional remote sensing object detection methods mainly rely on single-modal images, such as visible (RGB) or infrared (IR) images [4], [5], [6], [7]. However, single-modal detection faces many inherent limitations in complex remote sensing scenarios. For instance, the quality of RGB images is heavily dependent on lighting conditions [8]; in situations like nighttime, fog, or shadow occlusion, the texture and color features of objects degrade significantly, causing detection performance to drop sharply [9]. Although IR images have the advantage of all-day operation [10], their inherent low resolution and lack of texture detail often lead to false alarms and missed detections in densely packed or complex backgrounds [11]. In recent years, driven by advancements in multimodal fusion and deep learning [12], [13], [14], [15], [16], [17], multimodal remote sensing object detection has emerged as a key development direction in this field [18], [19], [20], [21]. It provides an effective approach to overcoming the limitations of single-modal detection.

However, the weakly misalignment [22] problem remains a core challenge in multimodal remote sensing object detection. This phenomenon arises because disparate sensor intrinsics and spatiotemporal synchronization errors render the precise alignment of captured images impossible. Such misalignment disrupts the spatial consistency of an object across modalities, severely degrading localization accuracy.

As illustrated in Fig. 1, the weakly misalignment problem in remote sensing involves significant *regional differences*, where the magnitude and direction of spatial offsets vary substantially across different local areas of the scene. Crucially, this phenomenon is particularly pronounced for *small objects*, as

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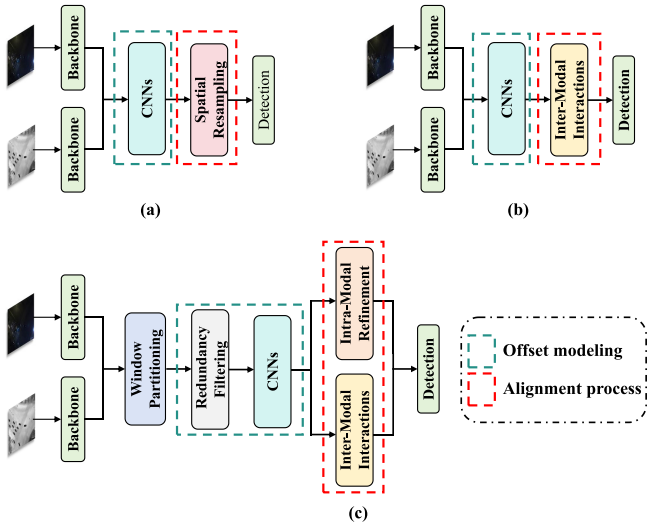


Fig. 2. Previous multimodal object detection architectures under weakly misalignment conditions versus our RPANet. (a) Explicit alignment method: typically employing CNNs to estimate spatial offsets for spatial resampling. (b) Implicit alignment method: leveraging cross-attention mechanisms to dynamically model intermodal relationships based on predicted offsets. (c) Our RPANet: introducing a progressive alignment strategy. We apply redundancy filtering before offset modeling, which then directs a synergistic implicit alignment process integrating intramodal refinement and intermodal interactions.

their limited spatial extent makes them highly sensitive to even minor pixel-level shifts [23].

To address the challenge of the weakly misalignment problem, existing methodologies generally fall into two primary categories: explicit alignment methods and implicit alignment methods, as shown in Fig. 2(a) and (b). Explicit alignment approaches [22], [24], [25], [26] seek to directly estimate spatial offsets between the feature maps of two modalities using convolutional neural networks (CNNs) and subsequently align the features through spatial resampling. However, due to the intrinsic locality of convolution operations, these approaches often ignore global contextual information that is crucial for remote sensing object detection [27]. Conversely, implicit alignment approaches [28], [29], [30], [31] leverage cross-attention mechanisms based on predicted offsets to dynamically compute feature similarity and adaptively align features, thereby capturing broader contextual information. However, relying solely on global intermodal interactions limits their ability to address the significant regional differences in misalignment. This is particularly detrimental to small objects, which require the precise, localized correction that global attention often fails to provide. Furthermore, their offset prediction is highly susceptible to interference from redundant information, and these methods frequently overlook the refinement and enhancement of salient features within each modality, as the fusion predominantly focuses on cross-modal relationships.

To overcome the aforementioned challenges, we propose a region-based progressive alignment network (RPANet) for multimodal remote sensing object detection, as shown in Fig. 2(c). Our framework addresses the critical issue of regional differences and small objects by decomposing the global alignment task into manageable local regions and performing progressive alignment from explicit offset prediction to implicit alignment. Specifically, the redundancy-aware regional offset modeling module (R^2OMM) first employs the regional spatial modulation (RSM) to partition the input features into local windows and enhance informative structures, and then performs channel-recalibrated offset modeling (COM) to suppress redundant responses and predict explicit spatial offsets. In addition, guided by the predicted offsets, the dual-path synergistic implicit alignment module (DPSIAM) utilizes a parallel synergistic attention block (PSAB) to jointly perform intramodal refinement and intermodal interaction, thereby achieving robust implicit alignment. In this way, RPANet effectively alleviates weak misalignment across varying local regions, especially for small objects.

The primary contributions of this work are outlined as follows.

1) To address the critical challenges of regional differences and small objects caused by weak misalignment in multimodal remote sensing, we propose RPANet. The proposed framework decomposes the global alignment task into local regions and performs progressive alignment in an explicit-to-implicit manner, thereby improving alignment adaptability to spatially varying displacement.
2) We propose R^2OMM for reliable local offset prediction. By integrating RSM and COM, this module enhances informative structures, suppresses redundant interference, and generates accurate explicit offsets for subsequent refinement.
3) We design DPSIAM for robust implicit alignment. Guided by the explicit offsets predicted by R^2OMM, this module leverages a parallel architecture of intramodal refinement and intermodal interaction to achieve effective implicit alignment.
4) Comprehensive experiments are conducted on multiple public datasets. Extensive evaluations on the DroneVehicle, VEDAI, DVTOD, and LLVIP benchmarks confirm that the proposed method substantially outperforms existing state-of-the-art approaches.

The subsequent structure of this article is organized as follows. Section II reviews the related work. Section III details the proposed methodology. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Work

A. Remote Sensing Object Detection

Contemporary object detection frameworks are typically divided into two primary paradigms: two-stage architectures, exemplified by Faster R-CNN [32], known for high accuracy, and single-stage models like YOLO [33] and RetinaNet [34], which are optimized for rapid inference to suit real-time scenarios. However, the direct application of these methods to remote sensing imagery is challenged by the arbitrary orientations of objects, which are poorly represented by standard horizontal bounding boxes (HBBs).

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To achieve superior localization fidelity, researchers have increasingly adopted oriented bounding boxes (OBBs) [35], [36]. For instance, Oriented R-CNN [35] adapts the two-stage Faster R-CNN [32] framework by incorporating a specialized network for generating oriented proposals. Meanwhile, S^2ANet [36] enhances one-stage detectors with a feature alignment module to generate high-quality, orientation-sensitive anchors. However, these single-modal approaches suffer from inherent limitations, as their reliance on a single source of information often leads to performance degradation in complex scenarios.

B. Multimodal Remote Sensing Object Detection

Multimodal remote sensing object detection aims to achieve robust, all-weather performance by fusing complementary information from RGB and IR sensors. Existing fusion strategies generally fall into three primary paradigms: pixel-level, decision-level, and feature-level fusion.

Pixel-level fusion operates at the input stage, merging raw images before they enter the neural network. Methods such as SuperYOLO [18] employ this strategy combined with auxiliary super-resolution branches to maintain low computational latency, though they often fail to fully exploit deep semantic interactions between modalities. Conversely, decision-level fusion occurs during the postprocessing phase, aggregating the predictions generated by separate detection streams. Approaches like UA-CMDet [9], and CRSIOD [37] utilize uncertainty-aware modules to weigh predictions from separate streams based on environmental factors, such as illumination conditions.

Feature-level fusion has become the dominant paradigm, typically utilizing two-stream backbones to extract modal-specific features and integrating them at various network depths. Within this category, substantial research has focused on managing the discrepancies between diverse data sources. Early works focused on disentangling modality-shared and modality-specific information; for instance, CMAFF [38] designs parallel selective and differential modules, while CALNet [39] and DDCINet [20] propose mechanisms to rectify semantic conflicts and filter irrelevant information. More recently, advanced architectures have been introduced to enhance fusion. Transformer-based methods [13], such as ICAFusion [40], leverage cross-attention for global modeling but often incur prohibitive computational costs. This has spurred the adoption of linear-complexity state space models (SSMs) [41]. For example, MGMF [21] and DMM [42] incorporate Mamba-based fusion modules to achieve efficient long-range dependency modeling.

Despite these advancements, existing methods frequently overlook a critical challenge inherent to remote sensing scenarios. They typically assume strict pixel-to-pixel correspondence between modalities. In practice, mechanical vibrations and optical parallax cause weak spatial misalignment, and direct fusion without alignment introduces feature noise and blurs object boundaries. Consequently, there is a pressing need for a fusion framework that can robustly handle weak misalignment to ensure accurate feature integration.

C. Alignment Learning for Multimodal Object Detection

A primary obstacle in multimodal object detection is the pervasive issue of the weakly misalignment caused by sensor variations. To address this challenge, alignment learning has emerged as a pivotal research direction aimed at correcting spatial disparities between modalities. Specifically, explicit alignment methods [22], [25], [26], [43], [44] seek to directly resolve spatial disparities by estimating spatial offsets between the feature maps of two modalities using CNNs. For instance, TSFADet [22] estimates global transformation parameters, while OAFA [26] and DPAL [25] predict pixel-wise shifts to warp features. To enhance the robustness of these predictions against modal discrepancies, COMO [43] utilizes high-level semantic features to guide the alignment, and RGFNet [44] employs illumination-invariant reflectance maps as stable references. However, due to the intrinsic locality of convolution operations, these approaches often ignore global contextual information that is crucial for remote sensing object detection, leading to suboptimal alignment in complex scenes.

Conversely, implicit alignment methods [28], [29], [30], [31] leverage cross-attention mechanisms based on predicted offsets to dynamically compute feature similarity and adaptively align features. Architectures such as C^2Former [28] and M-FPT [31] utilize intermodal cross-attention to calibrate features, while CF-Deformable DETR [30] relies on deformable sampling points. By utilizing cross-attention, these methods succeed in capturing broader contextual information. However, these global alignment strategies struggle to address the significant regional differences inherent in weak misalignment, where the magnitude and direction of deviation vary substantially across the scene. This issue is exacerbated for small objects due to their high susceptibility to even minute spatial deviations. Furthermore, the offset prediction in these methods is frequently corrupted by redundant cross-modal information, directly leading to inaccurate estimations. Finally, they frequently overlook the refinement of salient features within each modality, as the fusion process predominantly focuses on modeling cross-modal relationships.

To address these challenges, we propose the RPANet, which effectively resolves regional differences and enhances small object detection by decomposing the global alignment task into manageable local regions for progressive alignment.

III. Methodology

In this section, we present the detailed methodology of our proposed RPANet. We begin in Section III-A with an overview of the overall network architecture. Following this, we detail the two key submodules that comprise this block: the R^2OMM in Section III-B, and the DPSIAM in Section III-C. Finally, the loss function used for optimization is introduced in Section III-D.

A. Overview

As illustrated in Fig. 3, RPANet follows a dual-stream progressive alignment pipeline for RGB–IR object detection. Given an RGB image I_{rgb} and an IR image I_{ir}, two modality-specific backbones are first employed to extract multiscale

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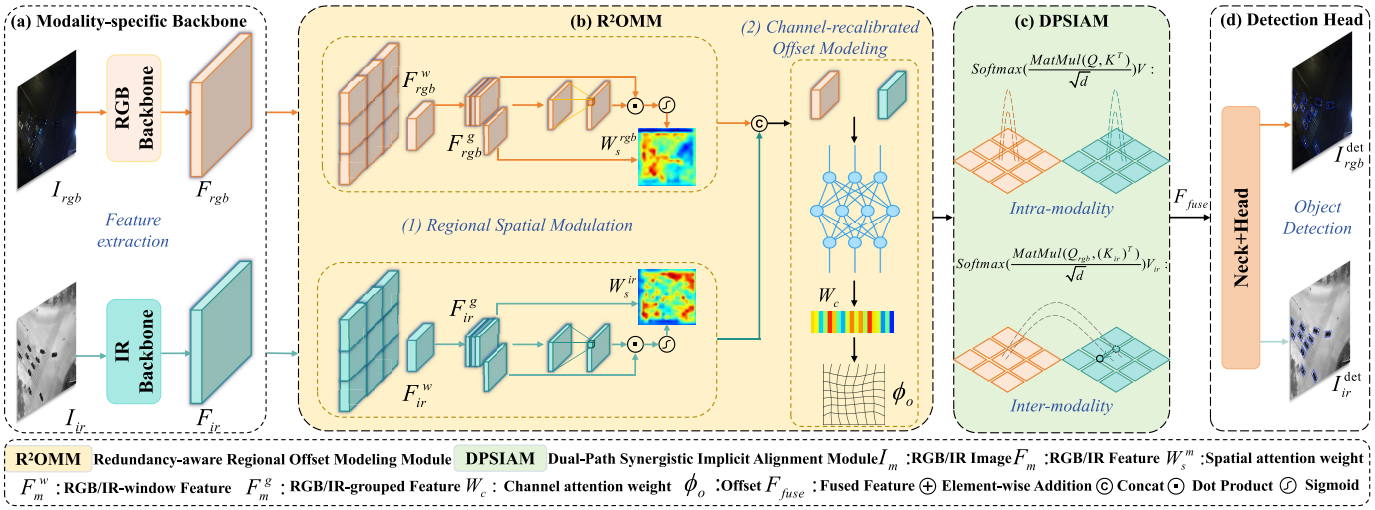


Fig. 3. Overall framework of the proposed RPANet. A dual modality-specific backbone extracts multiscale features from RGB and IR images. The extracted features are first processed by the R^{2}OMM, where the RSM suppresses regional spatial redundancy, and the COM further filters redundant cross-modal responses while predicting explicit regional offsets. Then, the DPSIAM performs offset-guided implicit alignment. Finally, the aligned features are fused for downstream object detection.

features from the two modalities. For simplicity, the RGB and IR features at one stage are denoted as F_{rgb} and F_{ir}, respectively. These paired features are then progressively processed by R^{2}OMM and DPSIAM before being fused for final detection.

In the first stage, R^{2}OMM performs explicit regional offset modeling. Specifically, the input RGB and IR features are divided into local windows, where RSM enhances informative spatial structures within each regional unit. Based on the spatially enhanced local features, COM further recalibrates cross-modal channel responses to suppress redundant information and generate explicit regional offsets. These offsets provide coarse geometric guidance for the subsequent alignment process.

In the second stage, DPSIAM performs offset-guided implicit alignment. The predicted offsets are used to sample the RGB features toward the IR reference space, while the IR features are kept as the reference. Then, DPSIAM jointly conducts intramodal refinement and intermodal interaction through a dual-path synergistic design. In this way, residual cross-modal mismatch can be further corrected while complementary information between RGB and IR features is enhanced.

Finally, the aligned RGB and IR features are concatenated and fused through a convolutional layer. The fused multiscale features from the last three stages are forwarded to the detection neck and head for object classification and localization.

B. Redundancy-Aware Regional Offset Modeling Module

Achieving reliable multimodal alignment fundamentally depends on the quality of explicit offset estimation [25]. However, weak misalignment in multimodal remote sensing images often exhibits evident regional variation, making a single global offset estimation strategy insufficient to capture locally inconsistent displacement patterns. This issue is particularly severe for small objects, whose sparse and weak cross-modal correspondence cues are easily overwhelmed by large-scale background regions during global offset estimation. Therefore, instead of directly estimating offsets on the entire feature maps, we decompose the global alignment problem into a set of local window-based subproblems, where each window serves as a relatively homogeneous alignment unit and enables more stable regional offset modeling.

Meanwhile, offset estimation within each local region can still be affected by redundant task-irrelevant responses [26], such as dominant background activations, low-discriminative textures, and modality-inconsistent responses that do not provide reliable cross-modal correspondence cues. These responses may overwhelm object-related structures, especially for small targets, causing the offset predictor to focus on irrelevant regions and produce unstable displacement estimates. Therefore, suppressing such redundancy is important for reliable regional offset modeling.

To address this, we propose the R^{2}OMM. As illustrated in Fig. 4, R^{2}OMM consists of two sequential components: the RSM, designed to establish regional alignment units and enhance spatially informative structures; and the COM, developed to suppress redundant responses through cross-modal channel recalibration and directly predict explicit regional offsets.

1) Regional Spatial Modulation: To reduce the difficulty of global misalignment estimation, we first partition the RGB and IR feature maps into local windows using alternating fixed window partition (FWP) and shifted window partition (SWP) [45]. FWP divides the feature maps into nonoverlapping windows of size $M \times M$, allowing offset estimation to be performed within compact local regions. However, fixed windows may isolate neighboring regions and weaken the modeling of objects or displacement patterns near window boundaries. Therefore, SWP shifts the partition by $(\lfloor M/2 \rfloor, \lfloor M/2 \rfloor)$, so that features originally belonging to adjacent FWP windows can be grouped into the same shifted window. By alternating FWP and SWP, RSM enables cross-window information exchange while preserving local alignment consistency, providing the

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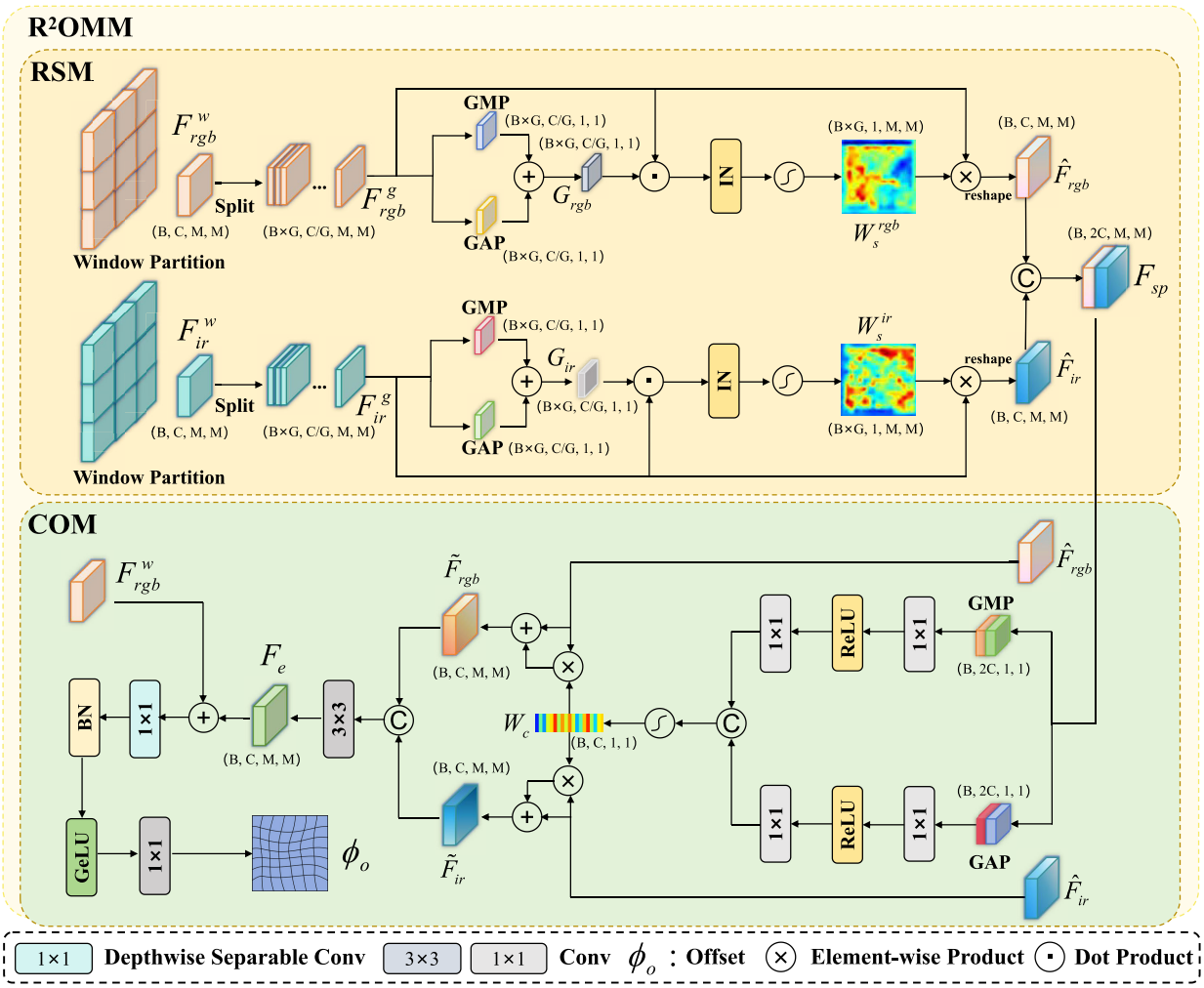


Fig. 4. Details of our proposed R²OMM. RSM stands for regional spatial modulation, and COM stands for channel-recalibrated offset modeling.

offset predictor with both local structural cues and neighboring contextual information for more stable regional offset modeling. In this way, we obtain window features $F^w_{rgb} \in \mathbb{R}^{B\times C\times M\times M}$ and $F^w_{ir} \in \mathbb{R}^{B\times C\times M\times M}$, where each window serves as a basic alignment unit.

To highlight spatially informative regions related to object structures, we then adopt a group-wise spatial modulation mechanism. Inspired by [46], for the input RGB and IR window feature maps F^w_{rgb} and F^w_{ir}, we split them into G groups along the channel dimension to obtain the grouped features $F^g_{rgb} \in \mathbb{R}^{(B\times G)\times C/G\times M\times M}$ and $F^g_{ir} \in \mathbb{R}^{(B\times G)\times C/G\times M\times M}$

$$F^g_{rgb} = Split\left(F^w_{rgb}\right) \tag{1}$$

$$F^g_{ir} = Split\left(F^w_{ir}\right). \tag{2}$$

To generate spatially adaptive attention weights, we process the grouped features through a dual-pooling strategy that integrates both global average pooling (GAP) and global max pooling (GMP). Then the outputs are added to form global statistics G_{rgb} and G_{ir}. Subsequently, we perform a dot product between these global statistics and the grouped features. This operation effectively scans the spatial extent, identifying regions that exhibit high semantic correlation with the group's global representation. To ensure numerical stability and highlight relative importance, the resulting similarity map is standardized using InstanceNorm (IN) [47]. Finally, a sigmoid function is applied to produce the final spatial attention weights W^{rgb}_s and W^{ir}_s, which can be formulated as

$$G_{rgb} = GAP\left(F^g_{rgb}\right) + GMP\left(F^g_{rgb}\right) \tag{3}$$

$$G_{ir} = GAP\left(F^g_{ir}\right) + GMP\left(F^g_{ir}\right) \tag{4}$$

$$W^{rgb}_s = \sigma\left(IN\left(G_{rgb}\cdot F^g_{rgb}\right)\right) \tag{5}$$

$$W^{ir}_s = \sigma\left(IN\left(G_{ir}\cdot F^g_{ir}\right)\right) \tag{6}$$

where σ represents the sigmoid function.

We multiply the generated spatial attention weights with the RGB and IR inputs, respectively, to obtain spatially significant features $\hat{F}_{rgb}$ and $\hat{F}_{ir}$. After concatenation, these features are combined to get cross-modal spatially enhanced features F_{sp}

$$\hat{F}_{rgb} = W^{rgb}_s \times F^g_{rgb} \tag{7}$$

$$\hat{F}_{ir} = W^{ir}_s \times F^g_{ir} \tag{8}$$

$$F_{sp} = Concat\left(\hat{F}_{rgb}, \hat{F}_{ir}\right). \tag{9}$$

2) Channel-Recalibrated Offset Modeling: Based on the spatially enhanced features, we further perform COM to suppress redundant responses and predict explicit regional offsets

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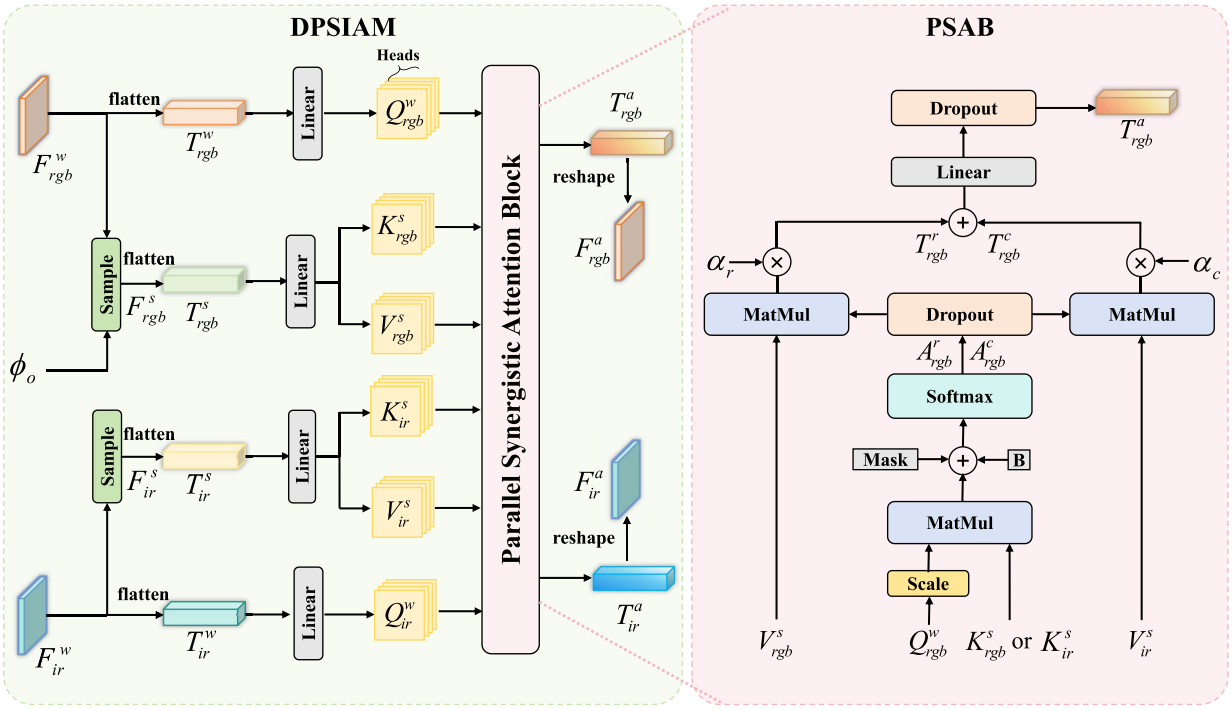


Fig. 5. Details of our proposed DPSIAM. For the PSAB, taking RGB as an example.

in a unified manner. Specifically, a cross-modal channel recalibration mechanism is first applied to model complementary interchannel relationships, enabling the network to adaptively emphasize informative features while effectively suppressing channel noise. The formula for channel attention weights W_c is as follows:

$$W_c = \sigma\left(Concat\left(Mlp\left(GAP\left(F_{sp}\right)\right)\right.\right.\\ \left.\left.Mlp\left(GMP\left(F_{sp}\right)\right)\right)\right) \tag{10}$$

where $Mlp(\cdot)$ consists of two 1×1 convolutions and a ReLU activation function. The channel weights are applied simultaneously to the features of both modalities, and the original features are enhanced through residual connections to get channel-enhanced features $\tilde{F}_{rgb}$ and $\tilde{F}_{ir}$

$$\tilde{F}_{rgb} = W_c \times \hat{F}_{rgb} + \hat{F}_{rgb} \tag{11}$$

$$\tilde{F}_{ir} = W_c \times \hat{F}_{ir} + \hat{F}_{ir}. \tag{12}$$

Finally, through concatenation and convolution operations, spatio-channel refined features F_e are generated. These features represent important spatial and channel information shared across modalities

$$F_e = Conv_{3\times3}\left(Concat\left(\tilde{F}_{rgb}, \tilde{F}_{ir}\right)\right). \tag{13}$$

After obtaining the spatio-channel refined features F_e, inspired by [48], we use a lightweight offset network that takes the RGB feature map F^w_{rgb} and the spatio-channel refined features F_e as input. This network generates corresponding offsets for each reference point of the original feature maps.

To predict the offsets, we first combine F_e with F^w_{rgb} and split the resulting feature map into G_o groups along the channel dimension. Each group is then processed by a shared lightweight offset prediction subnetwork to generate the corresponding offsets for the reference points with low computational cost. The explicit offsets $\phi_o \in \mathbb{R}^{(B\times G_o)\times 2\times M\times M}$ are formulated as

$$\phi_o = Conv_{1\times1}\left(GeLU\left(BN\left(DS_{1\times1}\left(F^w_{rgb} + F_e\right)\right)\right)\right) \tag{14}$$

where $DS_{1\times1}(\cdot)$ denotes the 1×1 depthwise separable convolution.

C. Dual-Path Synergistic Implicit Alignment Module

Cross-attention [49], [50], [51], [52] mechanisms are inherently adept at capturing broader contextual information and performing adaptive alignment by weighting interactions based on learned similarity scores [28]. However, existing methods typically rely exclusively on intermodal interactions, often overlooking the critical importance of intramodal refinement. Without intramodal refinement, the alignment process is prone to amplifying modality-specific noise, such as illumination variations in RGB images and thermal blurring in IR images. To address this limitation, we introduce DPSIAM. By synergizing intermodal interactions with intramodal refinement under the guidance of offset information, DPSIAM adaptively achieves robust and context-aware implicit alignment. The structure of our proposed DPSIAM is illustrated in Fig. 5.

Specifically, after obtaining the explicit regional offsets ϕ_o, we warp the feature map using a deformable grid sampling process. First, normalized reference sampling points $p \in [0, 1]^2$ are generated for each spatial position on the feature grid. The predicted offsets ϕ_o are then directly added to these reference points to form the actual sampling locations $p' = p + \phi_o$. Next, using these deformed sampling locations p', we apply bilinear interpolation as the sampling function $\mathcal{S}(\cdot)$ on the original RGB feature map F^w_{rgb} to obtain the sampled RGB

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feature map F^s_{rgb}. An analogous process is applied to the IR feature map F^w_{ir} to generate the sampled IR feature map F^s_{ir}

$$F^s_{rgb} = \mathcal{S}\left(F^w_{rgb}, p_{rgb} + \phi_o\right) \tag{15}$$

$$F^s_{ir} = \mathcal{S}\left(F^w_{ir}, p_{ir}\right). \tag{16}$$

Here, the offsets are applied only to the RGB branch because the IR modality is used as the annotation reference and generally provides more stable target responses under low-light conditions, while correcting RGB features toward the IR reference space also avoids reference ambiguity caused by simultaneously warping both modalities.

Then, to model the remaining inconsistency after explicit correction, we adopt a four-branch structure for implicit alignment between RGB features and IR features. First, we flatten the input original window features F^w_{rgb} and F^w_{ir} to obtain T^w_{rgb} and T^w_{ir}, and flatten the sampled RGB features F^s_{rgb} and the IR features F^s_{ir} to obtain T^s_{rgb} and T^s_{ir}. The queries Q^w_{rgb} and Q^w_{ir} are derived from the original window features through linear transformation, while the keys and values are obtained from the sampled feature maps via linear transformation to get K^s_{rgb}, K^s_{ir}, V^s_{rgb}, and V^s_{ir}

$$Q^w_{rgb} = T^w_{rgb}W^Q, Q^w_{ir} = T^w_{ir}W^Q \tag{17}$$

$$K^s_{rgb} = T^s_{rgb}W^K, K^s_{ir} = T^s_{ir}W^K \tag{18}$$

$$V^s_{rgb} = T^s_{rgb}W^V, V^s_{ir} = T^s_{ir}W^V \tag{19}$$

where W^Q, W^K, and $W^V \in \mathbb{R}^{C\times C}$ denote the weight matrices in linear layers.

To refine features within each modality and establish relationships between them, we design the PSAB, which employs a dual-stream architecture to concurrently compute intramodal self-refinement and intermodal cross correlation for each modality. Taking the RGB modality as an example, the intramodal feature refinement matrix A^r_{rgb} and the intermodal feature correlation matrix A^c_{rgb} are calculated as follows:

$$A^r_{rgb} = Softmax\left(\frac{MatMul\left(Q^w_{rgb}, \left(K^s_{rgb}\right)^T\right)}{\sqrt{d}} + B + Mask\right) \tag{20}$$

$$A^c_{rgb} = Softmax\left(\frac{MatMul\left(Q^w_{rgb}, \left(K^s_{ir}\right)^T\right)}{\sqrt{d}} + B + Mask\right) \tag{21}$$

where $MatMul$ denotes matrix multiplication, d refers to the dimension of each attention head, and B is the relative position encoding bias term used to represent the spatial position information within local windows. The softmax function normalizes the scaled correlation scores into a probability distribution, yielding the final attention weights. $Mask$ denotes the shifted window mask, similar to that used in Swin Transformer [45], preventing tokens from nonadjacent original windows from attending to each other after shifting.

After applying dropout to A^r_{rgb} and A^c_{rgb}, we multiply them by V^s_{rgb} and V^s_{ir}, respectively, to obtain the intramodal refined features T^r_{rgb} and intermodal interaction features T^c_{rgb}. To fully exploit the advantages of both types of features, we use a weighted fusion strategy to integrate them, yielding the final aligned feature sequences T^a_{rgb}

$$T^r_{rgb} = MatMul\left(Dropout\left(A^r_{rgb}\right), V^s_{rgb}\right) \tag{22}$$

$$T^c_{rgb} = MatMul\left(Dropout\left(A^c_{rgb}\right), V^s_{ir}\right) \tag{23}$$

$$T^a_{rgb} = Dropout\left(\left(\alpha_r T^r_{rgb} + \alpha_c T^c_{rgb}\right) W^O\right) \tag{24}$$

where α_r and α_c denote the learnable balancing coefficients subject to the constraint $\alpha_r + \alpha_c = 1$. Both are initialized to 0.5 to ensure an equal initial contribution. $W^O \in \mathbb{R}^{C\times C}$ refers to the output weight matrix. Finally, we reshape T^a_{rgb} back to its original spatial dimensions to obtain the aligned features F^a_{rgb}, and F^a_{ir} is computed in an analogous manner.

Algorithm 1 Algorithm of R^{2}OMM and DPSIAM

Input: $F^w_{rgb} : (B, C, M, M)$, $F^w_{ir} : (B, C, M, M)$

Output: $F^a_{rgb} : (B, C, M, M)$, $F^a_{ir} : (B, C, M, M)$

/* F^w_{rgb} and F^w_{ir} denote the RGB and IR window features, F^a_{rgb} and F^a_{ir} denote the RGB and IR aligned features. */\;

/* Redundancy-aware Regional Offset Modeling Module */

/* Regional Spatial Modulation */

$m \in \{rgb, ir\}$

$F^g_m : (B \times G, C/G, M, M) \leftarrow Split(F^w_m)$

$W^m_s : (B \times G, 1, M, M) \leftarrow Sigmoid(IN((GAP(F^g_m) + GMP(F^g_m)) \cdot F^g_m))$

$F_{sp} : (B, 2C, M, M) \leftarrow Concat(W^{rgb}_s \times F^g_{rgb}, W^{ir}_s \times F^g_{ir})$

/* Channel-recalibrated Offset Modeling */

$W_c : (B, C, 1, 1) \leftarrow \sigma(Concat(Mlp(GAP(F_{sp})), Mlp(GMP(F_{sp}))))$

$\tilde{F}_m : (B, C, M, M) \leftarrow W_c \times \hat{F}_m + \hat{F}_m$

$F_e : (B, C, M, M) \leftarrow Conv_{3\times3}(Concat(\tilde{F}_{rgb}, \tilde{F}_{ir}))$

$\phi_o : (B \times G_o, 2, M, M) \leftarrow Conv_{1\times1}(GeLU(BN(DS_{1\times1}(F^w_{rgb} + F_e))))$

/* Dual-Path Synergistic Implicit Alignment Module */

$F^s_{rgb} : (B, C, M, M) \leftarrow S(F^w_{rgb}, p_{rgb} + \phi_o)$

$F^s_{ir} : (B, C, M, M) \leftarrow S(F^w_{ir}, p_{ir})$

$F^a_m : (B, C, M, M) \leftarrow PSAB(F^w_{rgb}, F^w_{ir}, F^s_{rgb}, F^s_{ir})$\;

return F^a_{rgb}, F^a_{ir}\;

The detailed description of R^{2}OMM and DPSIAM can be seen in Algorithm 1.

D. Loss Function

Following the implicit alignment, the aligned features F^a_{rgb} and F^a_{ir} from the RGB and IR branches are fused via a convolutional layer to generate the fused features F_{fuse}. Specifically, the fused features from the last three stages are input into the detection neck and head to perform multiscale object detection. To optimize the proposed RPANet, the total objective function $\mathcal{L}_{total}$ is defined as

$$\mathcal{L}_{total} = \lambda_{box}\mathcal{L}_{box} + \lambda_{cls}\mathcal{L}_{cls} + \lambda_{dfl}\mathcal{L}_{dfl} \tag{25}$$

where $\mathcal{L}_{box}$, $\mathcal{L}_{cls}$, and $\mathcal{L}_{dfl}$ denote the bounding box regression loss, classification loss, and distribution focal loss, respectively, while λ_{box}, λ_{cls}, and λ_{dfl} serve as the balancing

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coefficients. In detail, $\mathcal{L}_{box}$ utilizes the ProbIoU [53] loss for OBB detection, and the CIoU loss for HBB detection. $\mathcal{L}_{cls}$ employs the binary cross-entropy (BCE) loss for accurate category prediction, and $\mathcal{L}_{dfl}$ is incorporated to further refine the bounding box regression by modeling the general distribution of box boundaries. Since R^2OMM and DPSIAM are differentiable, detection gradients can be directly propagated to the offset prediction and alignment processes, enabling task-oriented alignment learning without requiring dense cross-modal offset annotations.

IV. Experiments

A. Datasets and Evaluation Metric

We train and validate our model on four public multimodal object detection datasets.

DroneVehicle Dataset: The DroneVehicle [9] dataset is a large-scale benchmark for multimodal remote sensing object detection, designed to address the challenges of all-weather vehicle detection. It contains 28 439 paired RGB and IR images. The dataset includes 953 087 instances across five vehicle categories (car, bus, truck, van, and freight car), all annotated with OBB. It covers diverse day-to-night scenarios such as urban roads and parking lots. For standardized evaluation, the dataset is partitioned into training (17 990 pairs), validation (1469 pairs), and testing (8980 pairs) sets.

VEDAI Dataset: The VEDAI [72] dataset is a remote sensing detection benchmark containing a large number of small targets, featuring over 1200 high-resolution aerial RGB and IR image pairs. With image resolutions up to 1024×1024 pixels, the dataset contains more than 3700 annotated instances across diverse backgrounds such as urban roads, mountains, and farmland. It includes eight vehicle categories—such as car, truck, and van—with all objects originally annotated using OBB. Performance is evaluated following a standard tenfold cross-validation protocol.

DVTOD Dataset: The DVTOD [73] dataset is a benchmark designed for multimodal remote sensing object detection under cross-modal misalignment conditions. It contains 2179 RGB–IR image pairs (4358 images in total), with resolutions of 1920×1080 for RGB images and 640×512 for IR images. The dataset includes 6142 annotated instances using axis-aligned bounding boxes. It covers 54 capture scenes and 16 challenging attributes, including illumination variation, severe weather, object diversity, and occlusion.

LLVIP Dataset: The LLVIP [74] dataset is a visible-infrared paired benchmark designed for low-light vision and pedestrian detection. It contains 15 488 aligned RGB–IR image pairs, corresponding to 30 976 images in total. Most images are captured under very dark scenes, where visible images suffer from severe illumination degradation, while infrared images provide more reliable target cues. The dataset provides pedestrian annotations.

Evaluation Metric: To evaluate model performance, we use the standard mean average precision (mAP) metric. Our primary indicator is mAP@0.5, which averages the average precision (AP) across all categories using an intersection over union (IoU) threshold of 0.5 to define a true positive. The AP for each class is calculated from the area under its precision-recall curve, providing a comprehensive measure of detection performance across various confidence scores. In addition, we report mAP@0.5:0.95 on the DroneVehicle dataset to provide a more stringent evaluation of localization quality. This metric averages mAP over multiple IoU thresholds from 0.5 to 0.95 with a step size of 0.05, and therefore better reflects the detector's robustness under stricter localization requirements.

B. Implementation Details

All experiments were implemented based on the PyTorch 1.10.0 framework with CUDA 11.3, and conducted on a standalone NVIDIA GeForce RTX 3090 GPU. The proposed dual-stream architecture utilizes backbones initialized with MS COCO pretrained YOLOv8s weights. Optimization relied on stochastic gradient descent (SGD) with momentum set to 0.937 and weight decay to 0.0005. The learning rate followed a cosine annealing schedule, starting at 0.01 and reducing to 0.0001. For the DroneVehicle dataset, we trained the model for 100 epochs using a 640×640 input size and a batch size of 16. For the VEDAI dataset, the model was trained for 150 epochs with a 1024×1024 input resolution and a batch size of 8. For the DVTOD dataset, the model was trained for 150 epochs with an input resolution of 640×640 and a batch size of 16. For the LLVIP dataset, the model was trained for 300 epochs with an input resolution of 640×640 and a batch size of 16. Consistent with previous methodologies, the IR modality annotations were used as the ground truth for all experiments to ensure a fair comparison.

C. Comparison With State-of-the-Art Methods

1) Comparisons on DroneVehicle Dataset: To comprehensively evaluate the performance of our proposed RPANet, we conduct extensive comparisons on the challenging DroneVehicle dataset against a wide range of state-of-the-art methods. As shown in Table I, our model is benchmarked against six single-modal detectors, which include both one-stage (e.g., S^2ANet [36], RetinaNet [34], YOLOv5 [54], and YOLOv8 [55]) and two-stage (e.g., Faster R-CNN [32], Oriented R-CNN [35]) methods. In addition, we compare against thirteen advanced multimodal object detection methods, including UA-CMDet [9], TSFADet [22], C^2Former [28], CAGTDet [56], DDCINet [20], DMM [42], CALNet [39], CCLDet [58], Halfway Fusion [57], RGFNet [44], OAFA [26], ADMPF [59], and MDPMFN [60], which also represent a comprehensive mix of one-stage and two-stage methods. It clearly demonstrates that multimodal fusion methods generally outperform single-modal detectors.

The quantitative results in Table I demonstrate the superiority of our approach. RPANet achieves an mAP@0.5 of 82.35%, establishing a new SOTA on this benchmark and outperforming all other methods. This result surpasses the previous SOTA approach ADMPF [59] by 0.30% mAP@0.5. It is worth noting that our method surpasses the Mamba-based

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TABLE I

QUANTITATIVE BENCHMARKING AGAINST LEADING METHODS ON THE DRONEVEHICLE DATASET. WE BENCHMARK OUR RPANET AGAINST BOTH SINGLE-MODAL AND MULTIMODAL DETECTORS, ALL OF WHICH UTILIZE OBB DETECTION HEADS. THE FIRST AND SECOND RANKS ARE DISTINGUISHED BY BOLD AND UNDERLINING, RESPECTIVELY

Method	Backbone	Type	Modality	Car	Truck	Freight-car	Bus	Van	mAP@0.5	mAP@0.5:0.95
Faster R-CNN [32]	ResNet50	Two-Stage	RGB	77.10	46.00	34.50	76.90	34.20	53.70	26.06
RetinaNet [34]	ResNet50	One-Stage		75.20	42.50	30.40	74.00	26.20	49.70	21.97
YOLOv5 [54]	CSPDarkNet	One-Stage		88.84	64.32	42.10	87.38	44.18	65.36	37.42
S^2ANet [36]	ResNet50	One-Stage		77.30	49.80	38.20	78.00	39.30	56.50	22.89
Oriented R-CNN [35]	ResNet50	Two-Stage		77.70	50.10	38.40	77.90	37.40	56.30	29.61
YOLOv8 [55]	CSPDarkNet	One-Stage		93.22	72.73	51.22	93.62	52.73	72.70	43.81
Faster R-CNN [32]	ResNet50	Two-Stage	IR	90.00	59.60	45.10	87.20	40.20	64.40	37.53
RetinaNet [34]	ResNet50	One-Stage		89.20	52.80	39.80	85.20	33.20	60.00	36.47
YOLOv5 [54]	CSPDarkNet	One-Stage		80.10	69.35	56.62	88.05	50.30	70.89	40.41
S^2ANet [36]	ResNet50	One-Stage		90.10	63.90	51.40	88.30	44.20	67.60	39.39
Oriented R-CNN [35]	ResNet50	Two-Stage		90.10	63.80	52.40	88.50	41.40	67.20	42.89
YOLOv8 [55]	CSPDarkNet	One-Stage		97.84	77.51	66.92	94.58	57.80	78.73	48.92
UA-CMDet [9]	ResNet50	Two-Stage	RGB+IR	87.51	60.70	46.80	87.08	37.95	64.01	–
TSFADet [22]	ResNet50	Two-Stage		89.88	67.87	63.74	89.81	53.99	73.06	–
CALNet [39]	CSPDarkNet	One-Stage		90.30	76.15	62.97	89.11	58.46	75.39	–
C^2Former [28]	ResNet50	One-Stage		90.20	68.30	64.40	89.80	58.50	74.20	47.50
CAGTDet [56]	ResNet50	Two-Stage		90.82	69.65	66.28	90.46	55.62	74.57	–
Halfway Fusion [57]	ResNet50	Two-Stage		90.10	62.30	58.50	89.10	49.80	70.00	42.90
OAFA [26]	CSPDarkNet	One-Stage		90.30	76.80	**73.30**	90.30	66.00	79.40	–
DMM [42]	VMamba	One-Stage		90.40	79.80	68.20	89.90	**68.60**	79.40	**52.10**
CCLDet [58]	ResNet-FPN	Two-Stage		97.70	75.40	68.80	95.70	59.50	79.40	–
RGFNet [44]	CSPDarkNet	One-Stage		**98.40**	81.10	68.70	95.80	63.00	81.40	–
DDCINet [20]	ResNet50	Two-Stage		91.00	78.90	66.10	90.70	65.50	78.40	–
ADMPF [59]	CSPDarkNet	One-Stage		97.94	**82.28**	69.34	95.93	64.77	82.05	–
MDPMFN [60]	CSPDarkNet	One-Stage		97.95	80.87	67.68	95.50	64.33	81.26	–
RPANet (Ours)	CSPDarkNet	One-Stage		98.16	**82.28**	68.92	**96.32**	66.05	**82.35**	50.94

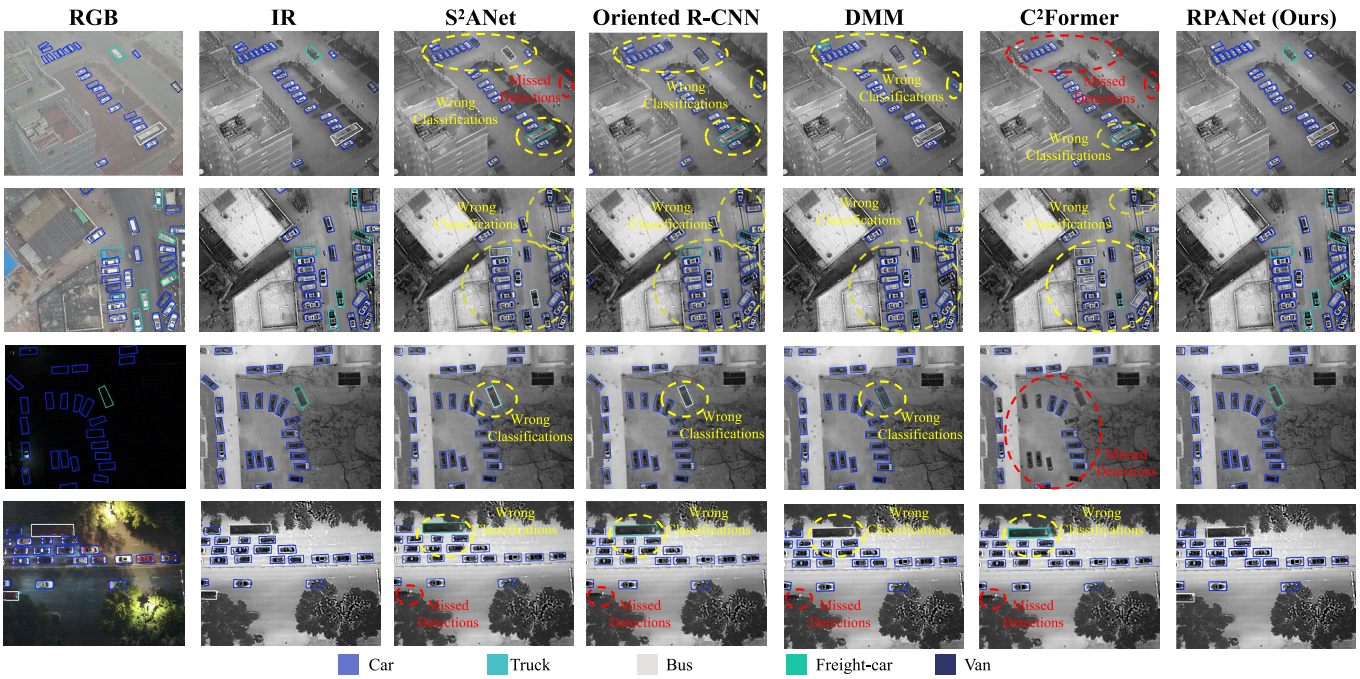


Fig. 6. Visualization of prediction results on the DroneVehicle dataset (confidence threshold: 0.5). The bounding boxes shown in the RGB and IR images denote the ground truth annotations. As highlighted by the red and yellow dashed regions, our proposed approach exhibits distinct advantages in detection accuracy compared to competing methods.

methods DMM [42] and RGFNet [44] by 2.95% and 0.95%, respectively, demonstrating its competitiveness against these latest architectures. Notably, RPANet achieves the highest AP score in the bus category and shares the best performance in the truck category, underscoring its superior capability in handling objects with varying scales. In addition to mAP@0.5, we further report the more stringent mAP@0.5:0.95 metric to provide a more comprehensive evaluation of localization quality. As shown in Table I, RPANet achieves 50.94% mAP@0.5:0.95, ranking second among the compared methods

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TABLE II

QUANTITATIVE BENCHMARKING AGAINST LEADING METHODS ON THE VEDAI DATASET. WE BENCHMARK OUR RPANET AGAINST BOTH SINGLE-MODAL AND MULTIMODAL DETECTORS, ALL OF WHICH UTILIZE OBB DETECTION HEADS. THE FIRST AND SECOND RANKS ARE DISTINGUISHED BY BOLD AND UNDERLINING, RESPECTIVELY

Method	Backbone	Type	Modality	Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	mAP@0.5
YOLOrs [61]	CSPDarkNet	One-Stage	RGB	85.25	72.93	70.31	50.65	42.67	76.77	18.65	38.92	57.00
CMAFF [38]	CSPDarkNet	One-Stage		91.10	82.30	75.10	78.30	33.30	81.20	71.80	62.20	71.90
SuperYOLO [18]	CSPDarkNet	One-Stage		90.30	82.66	76.69	68.55	53.86	79.48	58.08	70.30	72.49
S^2ANet [36]	ResNet50	One-Stage		88.10	82.60	83.30	74.30	47.60	65.80	73.20	72.90	73.50
Oriented R-CNN [35]	ResNet50	Two-Stage		87.40	85.80	**84.90**	69.10	39.50	68.90	58.40	56.40	68.80
YOLOrs [61]	CSPDarkNet	One-Stage	IR	82.03	73.92	63.80	54.21	43.99	54.39	21.97	43.38	54.71
CMAFF [38]	CSPDarkNet	One-Stage		86.70	75.90	66.60	77.10	43.00	62.30	70.70	84.30	70.80
SuperYOLO [18]	CSPDarkNet	One-Stage		87.90	81.39	76.90	61.56	39.39	60.56	46.08	71.00	65.60
S^2ANet [36]	ResNet50	One-Stage		85.50	82.30	78.20	76.90	25.10	57.00	49.90	78.00	66.60
Oriented R-CNN [35]	ResNet50	Two-Stage		86.00	84.10	81.90	76.10	33.80	56.00	65.70	75.50	69.90
YOLOrs [61]	CSPDarkNet	One-Stage	RGB+IR	84.15	78.27	68.81	52.60	46.75	67.88	21.47	57.91	59.73
CMAFF [38]	CSPDarkNet	One-Stage		91.70	85.90	78.90	78.10	54.70	71.90	71.70	75.20	75.90
SuperYOLO [18]	CSPDarkNet	One-Stage		91.13	85.66	79.30	70.18	57.33	80.41	60.24	76.50	75.09
CLGNet [62]	ResNet50	Two-Stage		-	-	-	-	-	-	-	-	80.20
C^2Former [28]	ResNet50	Two-Stage		87.20	80.70	82.70	77.40	58.40	72.90	71.40	75.20	75.70
ICAFusion [40]	CSPDarkNet	One-Stage		-	-	-	-	-	-	-	-	76.62
CrossYOLO [63]	CSPDarkNet	One-Stage		91.60	**90.60**	78.60	66.60	**74.00**	77.50	**81.50**	74.60	79.40
RPANet (Ours)	CSPDarkNet	One-Stage		**92.00**	85.60	77.00	**91.30**	65.90	**83.70**	62.10	**85.40**	**80.40**

with available results under this metric. In particular, RPANet exceeds C^2Former [28] and Halfway Fusion [57] by 3.44% and 8.04%, respectively. These results indicate that RPANet not only achieves the best detection accuracy under the commonly used IoU threshold of 0.5, but also maintains competitive localization performance under stricter IoU thresholds.

For a more direct performance perspective, we visualize the detection outputs of RPANet alongside other methods in Fig. 6, allowing a clear qualitative comparison. The examples highlight a variety of challenging scenarios prevalent in remote sensing, including foggy conditions, densely packed vehicles, and environments with extreme low-light. In these scenarios, it is evident that methods like the Mamba-based DMM [42] and the Transformer-based C^2Former [28] suffer from numerous missed detections and wrong classifications. In contrast, our RPANet successfully and accurately localizes and classifies these challenging instances. These visual comparisons substantiate the enhanced stability and localization precision of the developed framework.

2) Comparisons on VEDAI Dataset: To further validate the generalization capability of our method on small objects, we conduct comparative experiments on the VEDAI dataset. The quantitative results are presented in Table II. Our proposed RPANet is benchmarked against several single-modal and multimodal detectors, including YOLOrs [61], CMAFF [38], SuperYOLO [18], S^2ANet [36], Oriented R-CNN [35], C^2Former [28], ICAFusion [40], CLGNet [62], and CrossYOLO [63].

Our method achieves an mAP@0.5 of 80.40%, once again outperforming all competing methods. This result demonstrates a significant advantage over the best-performing single-modality detectors, exceeding the top RGB-only and IR-only models by 6.9% and 9.6%, respectively. Specifically, RPANet surpasses the second-best multimodal method, CLGNet [62], by 0.2%, and the third-best method, CrossYOLO [63], by 1.0%. In the per-category analysis, our method achieves the highest AP in challenging categories like truck, tractor, and van, and remains highly competitive in the car category.

TABLE III

QUANTITATIVE BENCHMARKING AGAINST LEADING METHODS ON THE DVTOD DATASET. WE BENCHMARK OUR RPANET AGAINST MULTIMODAL DETECTORS. THE FIRST AND SECOND RANKS ARE DISTINGUISHED BY BOLD AND UNDERLINING, RESPECTIVELY

Method	Backbone	Type	mAP@0.5
SuperYOLO [18]	CSPDarkNet	One-Stage	78.40
ICAFusion [40]	CSPDarkNet	One-Stage	81.00
GM-DETR [64]	HGNet [65]	One-Stage	80.10
QFDet [66]	ResNet50	Two-Stage	81.80
EI^2Det [67]	CSPDarkNet	One-Stage	81.30
RPANet (Ours)	CSPDarkNet	One-Stage	**82.40**

A qualitative comparison of detection results is provided in Fig. 7. These visualizations showcase complex urban, rural, and water-based remote sensing scenarios. The results show that while DMM [42] and C^2Former [28] exhibit missed detections and wrong classifications, our method accurately identifies these instances. This visually confirms the superior ability of our method to robustly identify small, multiclass objects within these complex remote sensing scenes.

3) Comparisons on DVTOD Dataset: To further evaluate the effectiveness of our method under cross-modal misalignment, we conduct comparative experiments on the DVTOD dataset, which is a drone-based visible–thermal benchmark captured under realistic misalignment conditions. The quantitative results are reported in Table III. Our proposed RPANet is compared against several recent multimodal detectors, including SuperYOLO [18], ICAFusion [40], GM-DETR [64], QFDet [66], and EI^2Det [67]. DVTOD is specifically designed to evaluate object detection under challenging misaligned scenarios.

As shown in Table III, our method achieves the best performance with an mAP@0.5 of 82.40%, outperforming all competing methods on this dataset. In particular, RPANet surpasses the second-best method, QFDet [66], by 0.6%, and improves over other strong one-stage detectors such as EI^2Det [67] and ICAFusion [40] by 1.1% and 1.4%, respectively. These results demonstrate that the proposed region-based

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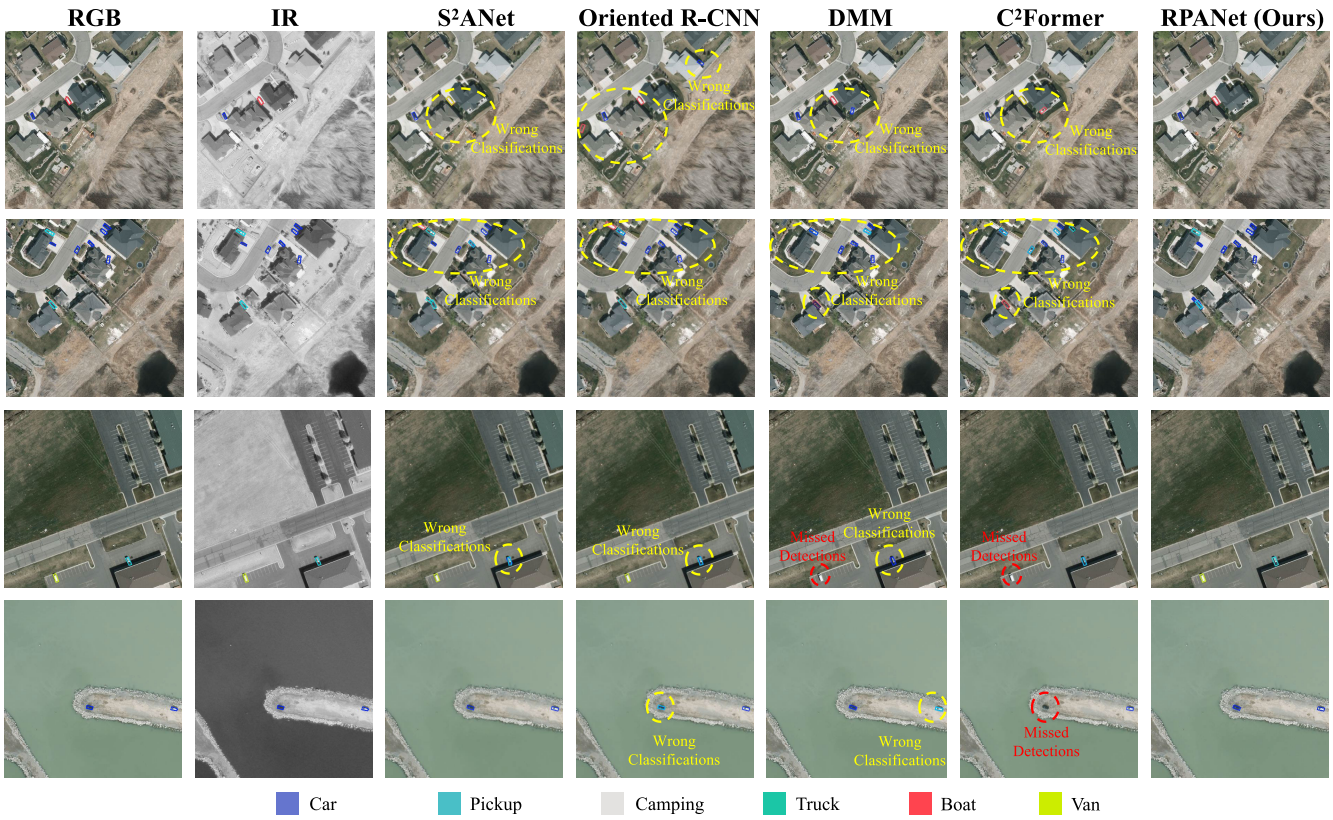


Fig. 7. Visualization of prediction results on the VEDAI dataset (confidence threshold: 0.5). The bounding boxes shown in the RGB and IR images denote the ground truth annotations. As highlighted by the red and yellow dashed regions, our proposed approach exhibits distinct advantages in small objects against complex backgrounds.

TABLE IV

QUANTITATIVE BENCHMARKING AGAINST LEADING METHODS ON THE LLVIP DATASET. WE BENCHMARK OUR RPANET AGAINST MULTIMODAL DETECTORS. THE FIRST AND SECOND RANKS ARE DISTINGUISHED BY BOLD AND UNDERLINING, RESPECTIVELY

Method	Backbone	Type	mAP@0.5
CSAA [68]	ResNet50	Two-Stage	94.30
ICAFusion [40]	CSPDarkNet	One-Stage	95.20
RSDet [69]	ResNet50	Two-Stage	95.80
ACDF-YOLO [70]	CSPDarkNet	One-Stage	96.50
MSFF [71]	CSPDarkNet	One-Stage	<u>96.80</u>
RPANet (Ours)	CSPDarkNet	One-Stage	**96.94**

TABLE V

COMPARISON OF COMPUTATIONAL EFFICIENCY AND PERFORMANCE FOR DIFFERENT DETECTION METHODS ON THE DRONEVEHICLE DATASET. THE FIRST AND SECOND RANKS ARE DISTINGUISHED BY BOLD AND UNDERLINING, RESPECTIVELY

Method	Params (M)	FLOPs (G)	FPS	mAP@0.5
TSFADet [22]	-	109.80	<u>18.60</u>	73.06
CAGTDet [56]	-	120.60	17.80	74.57
C^2Former [28]	132.50	<u>100.90</u>	-	74.20
RGFNet [44]	<u>79.80</u>	-	**46.20**	<u>81.40</u>
RPANet (Ours)	**54.81**	**59.72**	13.30	**82.35**

progressive alignment strategy is effective for handling the cross-modal misalignment and complex scene variations presented in DVTOD. In addition, the strong performance on this challenging dataset further validates the robustness of RPANet for multimodal object detection in realistic scenarios.

4) Comparisons on LLVIP Dataset: To further evaluate the generalization capability of our proposed RPANet beyond vehicle-oriented multimodal detection scenarios, we conduct comparative experiments on the LLVIP dataset, which focuses on visible–infrared pedestrian detection under low-light conditions. As shown in Table IV, RPANet is compared with several representative multimodal detectors, including CSAA [68], ICAFusion [40], RSDet [69], ACDF-YOLO [70], and MSFF [71].

The quantitative results show that our proposed RPANet achieves the best performance with an mAP@0.5 of 96.94%, outperforming all competing methods. In particular, RPANet surpasses the strong one-stage detector MSFF by 0.14% and ACDF-YOLO by 0.44%. These results demonstrate that the proposed region-based progressive alignment framework is not limited to vehicle detection, but can also generalize effectively to pedestrian detection in low-light visible–infrared scenarios. This further validates the robustness and applicability of RPANet across different multimodal detection categories.

5) Comparisons of Computational Costs: To evaluate the computational efficiency of our proposed RPANet, which is critical for deployment on resource-constrained platforms, we compare its model parameters (Params) and FLOPs against the latest alignment methods in Table V. Our method achieves an SOTA mAP@0.5 of 82.35% while maintaining the lowest

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TABLE VI
RESULTS OF THE ABLATION ANALYSIS OF THE KEY MODULES IN RPANET ON THE DRONEVEHICLE DATASET

No.	Modality	R^2OMM	DPSIAM	mAP@0.5
1	RGB	-	-	72.70
2	IR	-	-	78.73
3	RGB+IR	-	-	79.54
4	RGB+IR	✓	-	81.64
5	RGB+IR	-	✓	81.82
6	RGB+IR	✓	✓	**82.35**

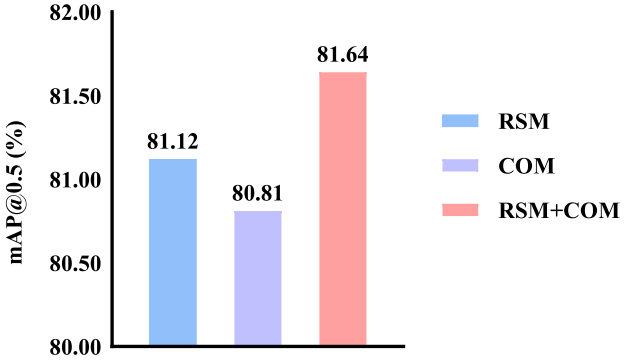


Fig. 8. Results of the ablation analysis of the R^2OMM on the DroneVehicle dataset.

Params and FLOPs among methods with available results, with only 54.81-M Params and 59.72G FLOPs. This is significantly more lightweight than other alignment methods like C^2Former [28], RGFNet [44], TSFADet [22], and CAGTDet [56]. The empirical data substantiate that RPANet achieves an ideal equilibrium between detection precision and computational economy, thereby establishing itself as a viable candidate for practical Earth observation tasks. In terms of inference speed, RPANet achieves 13.30 FPS, providing a practical reference for deployment efficiency while maintaining the best detection accuracy.

D. Ablation Study

We performed extensive ablation studies on the DroneVehicle benchmark to rigorously assess the specific impact of individual modules within the developed framework. Quantitative outcomes are presented in Table VI. The discussion begins with single-modality results, followed by an examination of the multimodal baseline and the effectiveness of the respective modules.

1) Baseline Setup: As shown in Table VI, the single-modality baselines for RGB (No. 1) and IR (No. 2) achieve 72.70% and 78.73% mAP@0.5, respectively. The superior performance of the IR-only model highlights the limitations of RGB-light sensors in complex lighting conditions (e.g., nighttime) and confirms the necessity of a robust multimodal fusion strategy. Our multimodal baseline (No. 3), which utilizes a dual-stream architecture based on YOLOv8s with a simple fusion for RGB + IR inputs, establishes a performance of 79.54% mAP@0.5. This result serves as the primary benchmark against which our proposed modules are evaluated.

2) Effectiveness of R^2OMM: R^2OMM is designed to improve regional offset estimation by decomposing the global alignment task into local windows, enhancing informative structures, and suppressing redundant responses before offset prediction. To verify its effectiveness, we investigate the contributions of RSM and COM. As shown in Fig. 8, when only RSM is adopted, the offset prediction branch is still retained, yielding 81.12% mAP@0.5. In contrast, using only COM achieves 80.81% mAP@0.5. When RSM and COM are jointly adopted, the performance further improves to 81.64% mAP@0.5. Compared with using RSM or COM alone, the complete R^2OMM brings gains of 0.52% and 0.83%, respectively. These results demonstrate that RSM and COM play complementary roles in regional offset modeling: RSM helps establish locally informative representations, while COM further suppresses redundant cross-modal interference and produces more reliable offsets. Their combination, therefore, provides a stronger basis for accurate offset prediction.

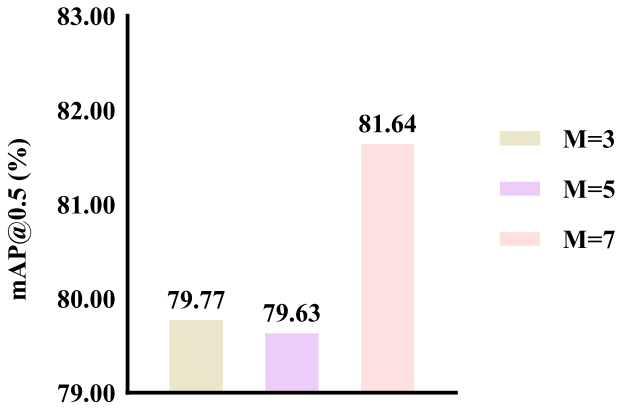


Fig. 9. Influence of the window size M on the DroneVehicle dataset.

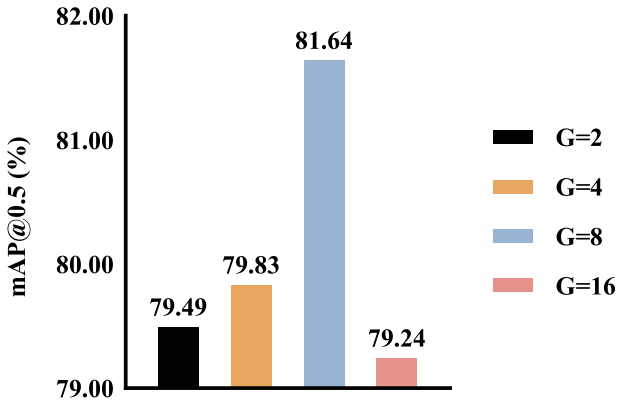


Fig. 10. Influence of the group number G on the DroneVehicle dataset.

In addition, we further analyze the influence of two key hyperparameters in R^2OMM, including the window size M and the group number G. As shown in Fig. 9, when M is set to 3, 5, and 7, the corresponding mAP@0.5 values are 79.77%, 79.63%, and 81.64%, respectively. The best performance is achieved when $M = 7$, indicating that a relatively larger local window can provide richer contextual information for regional offset modeling, while smaller windows may restrict the available spatial context and weaken the reliability of offset estimation. Moreover, we evaluate different group numbers in the group-wise spatial modulation. As shown in Fig. 10, the

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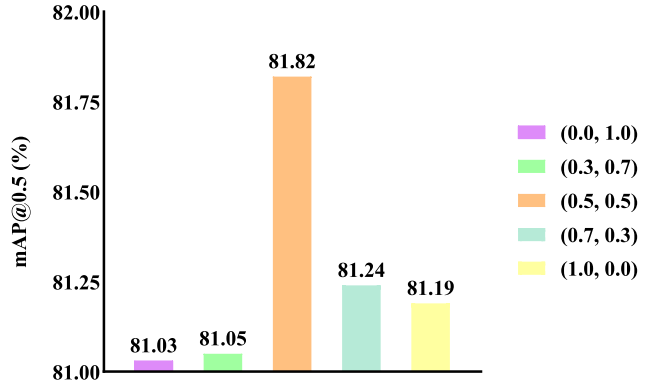


Fig. 11. Parametric study of the DPSIAM on the DroneVehicle dataset.

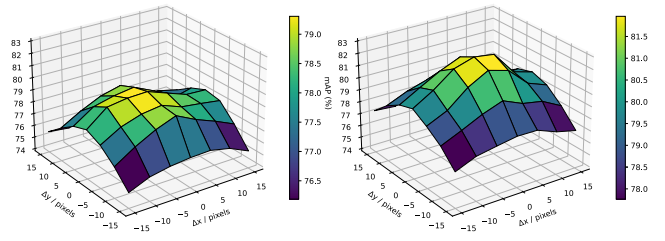


Fig. 12. Surface plots for the position shift analysis. The horizontal axes represent the imposed shifts along the x- and y-directions, while the vertical axis reports the corresponding detection accuracy in terms of mAP@0.5. The left plot represents the Baseline, and the right plot represents RPANet.

mAP@0.5 values are 79.49%, 79.83%, 81.64%, and 79.24% when G is set to 2, 4, 8, and 16, respectively. The best result is obtained at $G = 8$, suggesting that a moderate group number can better balance feature diversity and representation stability. When G is too small, the grouped features may lack sufficient diversity, while an excessively large G may fragment channel information and weaken feature representation. Therefore, we set $M = 7$ and $G = 8$ as the default settings in our experiments.

3) Effectiveness of DPSIAM: DPSIAM is introduced to perform implicit alignment based on the explicit offsets predicted by R^{2}OMM. Its effectiveness is validated by comparing the full model (No. 6) with a variant equipped with R^{2}OMM but without DPSIAM (No. 4). As shown in Table VI, introducing DPSIAM improves the mAP@0.5 from 81.64% to 82.35%, yielding a gain of 0.71%. This improvement demonstrates that, beyond explicit offset prediction, implicit refinement is still essential for correcting residual mismatch and enhancing cross-modal feature interaction.

To further investigate the optimal synergy within DPSIAM, we conduct a parametric study on the balancing coefficients α_r and α_c in Fig. 11. The results show that the model achieves the best performance when the two branches are equally weighted, i.e., $\alpha_r = 0.5$ and $\alpha_c = 0.5$. Deviating from this balanced setting leads to performance degradation. In particular, relying exclusively on intermodal interaction ($\alpha_r = 0.0$, $\alpha_c = 1.0$) or intramodal refinement ($\alpha_r = 1.0$, $\alpha_c = 0.0$) reduces the mAP@0.5 to 81.03% and 81.19%, respectively. These results confirm that neither intramodal refinement nor intermodal interaction alone is sufficient, while their synergistic integration is critical for robust implicit alignment.

4) Overall Effects: Our complete model, RPANet (No. 6), integrates all proposed components and achieves a final mAP@0.5 of 82.35%. This represents a total improvement of 2.81% over the dual-stream baseline (No. 3). The results in Table VI clearly demonstrate the effectiveness of our proposed modules when integrated into this complete, synergistic framework.

E. Performance Under Position Shift

To further evaluate the effect of weak misalignment, we conduct additional experiments on the DroneVehicle dataset by manually introducing spatial shifts between RGB and IR image pairs. Specifically, the model is trained on the original training set and evaluated on shifted image pairs, where the IR images are kept fixed, and the RGB images are shifted along the x-axis and y-axis with different magnitudes. The quantitative results are shown in Fig. 12.

As shown in Fig. 12, the detection performance of both the baseline and RPANet gradually decreases as the positional shift increases, indicating that larger cross-modal displacement still poses challenges for multimodal object detection. Nevertheless, RPANet consistently achieves better detection performance than the baseline across different shift settings. This result demonstrates that the proposed alignment strategy remains effective under varying degrees of positional shift.

F. Feature Analysis

To provide an intuitive validation of our proposed method, we visualize the intermediate feature maps from different model configurations on the DroneVehicle dataset, as shown in Fig. 13. This qualitative analysis aims to demonstrate the effectiveness of our modules in mitigating weak misalignment and suppressing background noise.

As clearly observed in Fig. 13, the baseline model's feature activations are suboptimal. In well-lit daytime scenes, the activations are diffuse and fail to precisely focus on the dense vehicle objects, leading to blurred boundaries between instances. In challenging nighttime and low-light scenarios, the feature response is extremely weak, rendering many objects nearly invisible on the feature map. This illustrates the severe impact of weak misalignment and background noise.

With the introduction of DPSIAM, the feature quality is significantly improved. Through implicit alignment, the model begins to locate objects more accurately, resulting in more focused activation regions and partial suppression of background noise. However, in the low-light scenes, while object saliency is enhanced, some activations still lack sufficient intensity.

In contrast, our full RPANet model (R^{2}OMM + DPSIAM) demonstrates superior feature representation. After introducing the R^{2}OMM, the model first obtains precise, redundancy-aware explicit offset guidance. This allows the subsequent implicit alignment by DPSIAM to operate with maximum effectiveness. As shown in the fifth column of Fig. 13, the activation regions become exceptionally sharp and focused, regardless of the scene—from dense daytime parking lots to nighttime roads. Object features are significantly enhanced, and background noise is almost entirely suppressed.

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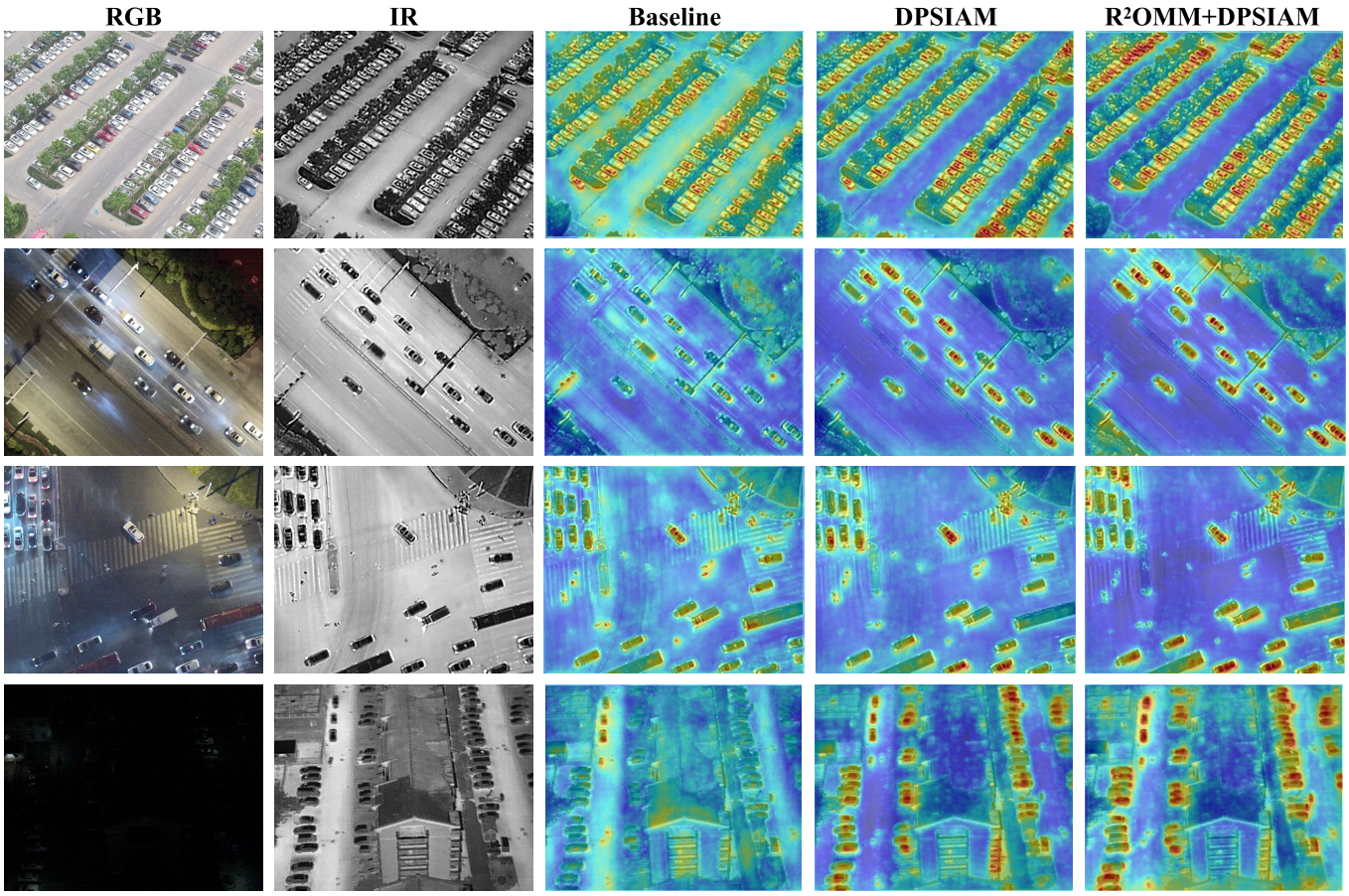


Fig. 13. Visualization of feature maps on the DroneVehicle dataset. (From left to right) Results are baseline, with DPSIAM added, and with both R^{2}OMM and DPSIAM added.

In summary, the visual results strongly support the synergistic effect of our progressive alignment strategy. The baseline suffers from diffuse activations due to misalignment, while our full RPANet successfully resolves the regional differences of misalignment by R^{2}OMM and DPSIAM.

V. Conclusion

In this article, we propose RPANet for multimodal remote sensing object detection under weakly misalignment. By decomposing the global alignment task into local regions and performing progressive explicit-to-implicit alignment, RPANet effectively handles spatially varying misalignment and improves the detection of small objects. Specifically, R^{2}OMM predicts regional explicit offsets by enhancing informative structures and suppressing redundant responses, while DPSIAM performs offset-guided implicit alignment through parallel intramodal refinement and intermodal interaction. Extensive experiments on several challenging public datasets demonstrate that RPANet achieves state-of-the-art performance with a favorable balance between accuracy and computational cost. Comprehensive ablation studies further validate the effectiveness of all proposed components. In the future, we will explore semantic-aware fusion strategies to further enhance multimodal feature integration. Specifically, semantic cues such as object categories, foreground regions, and scene context can be introduced to adaptively guide RGB–IR feature fusion, helping the model select more reliable modality information under complex illumination and background conditions.

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